



Exploring bonding temperature variations in diffusion brazing of B₄C ceramics with Ni-Cr-Si-Fe-B interlayer: Mechanical and microstructural analysis



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Abstract

This study examines the impact of the bonding temperature parameter on the quality of diffusion brazing joints of B₄C ceramics using a Ni-Cr-Si-Fe-B interlayer. Various factors including shear strength, hardness, microstructure, interfacial phenomena, elemental diffusion patterns, formed compounds and phases, and fracture surfaces of the samples were analyzed and compared. The brazed sample at 1110°C for 1/2h exhibited the highest shear strength of 62MPa. Increase of the bonding temperature to the optimum level (1090–1110°C) enhances the fluidity of the molten interlayer, facilitating its diffusion into the porosities of the ceramic components. This increase also promotes higher interactions within the system, creating favorable conditions for

element diffusion. However, surpassing the optimum temperature (1110–1130°C) may lead to increased destructive reactions and intensify the impact of the coefficient of thermal expansion (CTE) mismatch, resulting in residual stresses and a subsequent decrease in bond strength. The excessive increase in the bonding temperature (1130–1150°C) can lead to the excessive fluidity of the molten interlayer, potentially causing effusion onto the joint's fixture and damaging the samples. The results indicate the formation of various compounds between the elements of the interlayer and carbon and boron elements, including SiC, Ni₄B₃, CrB₂, Fe₂B, Ni₆Si₂B, and B₂Fe₃Ni₃, at the fracture surface of the sample bonded at 1110°C. The more substantial peaks of these compounds, compared to other samples, could be a reason for the higher strength of this particular sample.



Keywords

B₄C; Ni-Cr-Si-Fe-B interlayer; Diffusion brazing; Bonding temperature

1. Introduction

The exceptional chemical, mechanical, and physical properties of B₄C ceramic, including high hardness, excellent abrasive resistance, and good thermal resistance, make it a crucial non-oxide ceramic widely employed across various industries and technologies. Despite these desirable characteristics, the inherent brittleness of B₄C ceramic, along with challenges related to its joining and forming, imposes limitations on its applications. To overcome these limitations, researchers have focused on developing advanced joining techniques. The successful establishment of sound metallurgical bonds between B₄C parts and between this ceramic and other metals holds the potential to broaden its utility as a structural material in diverse industries and technologies [[1], [2], [3]]. One application of B₄C is in producing reinforced ceramic composites, which are remarkable materials with enhanced mechanical properties. Various methods exist for manufacturing these composite materials, many involving infiltrating a liquid matrix into a porous reinforcement medium. Reactive infiltration has proven to be one of the most efficient techniques, and numerous studies have been conducted to explore the effects of different parameters on optimizing the manufacturing of composite materials using this method [[4], [5], [6], [7], [8]]. While conventional fusion welding methods have

proven effective for joining metallic alloys [9,10], using these methods for joining most ceramics is challenging and will not lead to favorable results. Therefore, alternative techniques such as brazing and diffusion bonding have been extensively researched. Researchers have turned to advanced methods like brazing and diffusion bonding in recent years to explore novel avenues for joining ceramics [[11], [12], [13], [14], [15]].

Sekine et al. explored the low-temperature bonding of B₄C ceramic parts using an aluminum layer within a temperature range of 600–1200°C under a vacuum of 10^{-2} - 10^{-4} Pa for 2 h [16]. The results demonstrated successful bonding and high flexural strength, closely approaching that of B₄C ceramic, for samples bonded in the 700–1100°C temperature range. The scanning electron microscopy (SEM) observations revealed a dense middle layer and a crack-free interface, along with the diffusion of aluminum into B₄C surface micro-cracks. The electron probe microanalysis (EPMA) confirmed the formation of compounds such as AlB₂ and Al₃BC in the middle layer, contributing to developing a high-strength composite middle layer.

Vosughi et al. investigated the diffusion bonding of B₄C/B₄C using a copper interlayer [17]. The reasons for choosing copper are properties such as suitable ductility, low melting point, and high diffusion capability of this element. The results showed that by increasing the temperature and time, the measured thickness of the diffusion layer increased. The joint primarily formed due to the diffusion of copper into the porosities of the base ceramic body near the interface, with the highest reported shear strength of 170 MPa obtained for the bonded sample at 950°C for 45 min.

Wang et al. studied a Ni-B₄C couple prepared by macroscopic diffusion welding and a Ni-15wt%B₄C composite produced by the spark plasma sintering (SPS) method to understand micro-scale diffusion bonding between ceramics and metals [18]. In the Ni-B₄C couple, a complete diffusion welding joint has been achieved. The microscopic analysis of Ni-15wt%B₄C revealed loose structures around B₄C particles. The X-ray Diffraction (XRD) analysis indicated distinct stages in the diffusion bonding process during the SPS method. Overall, the diffusion bonding process in the two studied couples has been reported to be completely different.

Regarding the brazing of B₄C components using a 63Ag-35.25Cu-1.75Ti filler alloy, increasing bonding temperature, bonding time, and filler alloy thickness was found to enhance shear strength [11]. The highest reported strength was attained for a brazed sample at 870°C for 15 min with a filler alloy thickness of 220 µm (79 MPa).

Aizenshtein et al. examined the brazing joint of B₄C parts using three different copper-based filler alloys [19]. The joint with a copper-base filler alloy containing 9 at.% of boron exhibited the highest shear strength of 100 MPa. The absence of brittle phase

formation in the joint interface has been identified as the main contributing factor to this outcome.

This study is groundbreaking as it is the first to comprehensively combine mechanical tests with detailed microstructural investigations to evaluate the impact of bonding temperature on the quality of diffusion brazing joints in B₄C ceramics using a Ni-Cr-Si-Fe-B (BNi-2) interlayer (refer to [Table 1](#)). This dual approach offers a more holistic understanding of the bonding process and its outcomes. Additionally, the research highlights the precise temperature window where optimal shear strength and joint quality will be achieved, thus providing critical insights for industrial applications aiming to enhance joint performance. The study meticulously correlates mechanical properties such as shear strength and hardness with microstructural phenomena like elemental diffusion patterns and compound formations, which not only elucidates the underlying mechanisms but also facilitates the prediction and control of joint quality. The unique combination of analytical techniques alongside mechanical testing makes the findings more robust and comprehensive compared to previous studies.

Table 1. Chemical analysis of Ni-Cr-Si-Fe-B Interlayer (wt. %) [[20](#)].

Ni	Cr	B	Si	Fe	C	P	S	Al	Ti	Zr	Co	Se	Others
Bal.	6.0– 8.0	2.75– 3.50	4.0– 5.0	2.5– 3.5	0.06	0.02	0.02	0.05	0.05	0.05	0.10	0.005	0.50

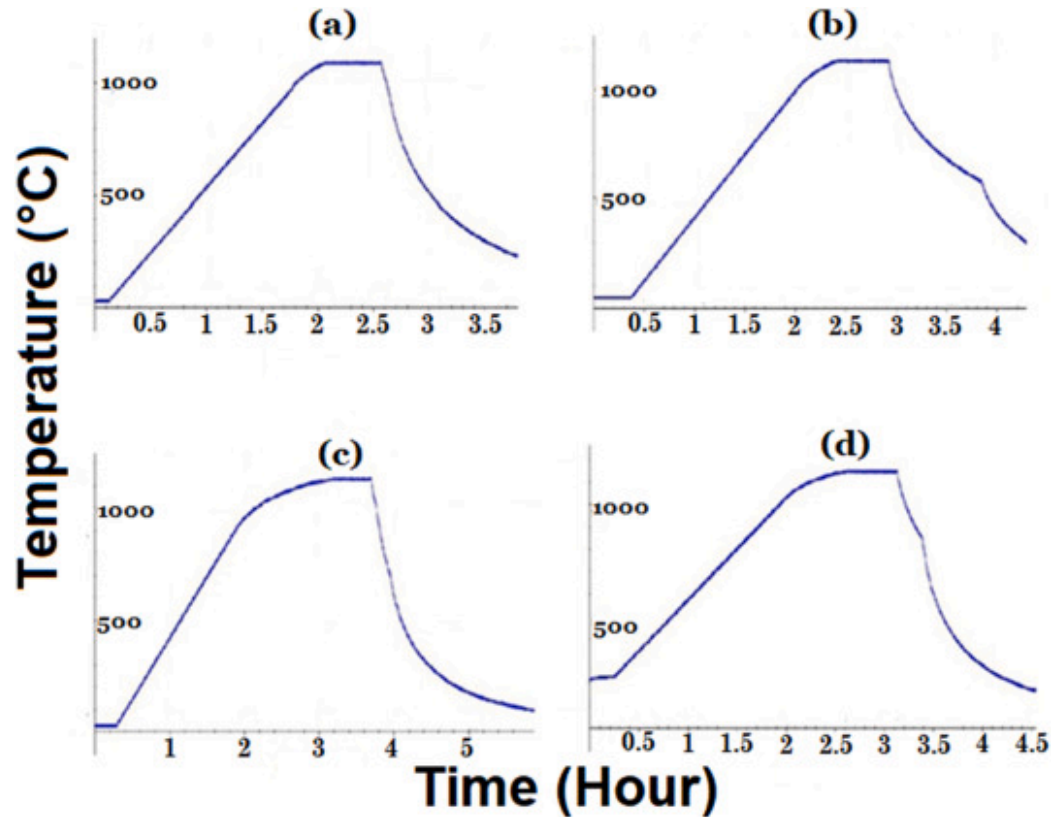
2. Experimental procedure

The ceramic pieces of B₄C were cut from the primary block of B₄C prepared using the hot press method with a density close to theoretical density. These cut pieces were then polished and prepared with a surface area of 10 * 10mm² (dimensions: Length=10mm, Width=10mm, Thickness=5mm). A cutting machine equipped with a diamond blade was employed for cutting the ceramic parts. The diamond blade of the machine is rotated by a simple rotor at an adjustable speed. The samples are held on the blade by a holder and cut.

A 60µm thick layer of BNi-2 interlayer was placed between two prepared B₄C ceramic pieces to create brazing samples. This assembly was then positioned in a brazing fixture and transferred to TEF-1200 furnace (Artine Sana'at Parto Co.) under a vacuum

of 10^{-4} torr. The design and operating temperatures of this furnace are 1200°C and 1170°C, respectively. The length of the hot zone in this furnace is 25 cm.

Brazing joints were performed at temperatures of 1090, 1110, 1130, and 1150°C (S1, S2, S3, and S4 samples, respectively) for a bonding time of 1/2h. The bonding time was chosen based on the previous results of B₄C wetting by the BNi-2 interlayer [21]. Fig. 1 illustrates the brazing cycle for each of the mentioned temperatures. For each condition, two samples (one for shear testing and one for microscopic analyses) were brazed simultaneously during the thermal cycle to minimize potential errors.



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Fig. 1. The performed brazing cycles for the connections at temperatures: a) 1090°C, b) 1110°C, c) 1130°C, and d) 1150°C.

Microscopic study, elemental analysis, and phase investigation of the joints were conducted using Scanning Electron Microscopy (SEM), Energy Dispersive X-Ray Spectroscopy (EDS), X-ray Diffraction (XRD), and Wavelength Dispersive X-Ray Spectroscopy (WDS) methods. SEM and EDS analyses were done by Tescan Mira3 field emission scanning electron microscope (FESEM). The phases were identified by a X-ray diffractometer (PANalytical X'Pert PRO, the Netherlands) equipped with a Cu K α source using PW3050/60 goniometer. The continuous scanning was carried out from 5.0131° as 2 θ start position to 79.9711° as 2 θ end position, with a step size of 0.0260° and a scan step time of 37.9950s. Additionally, shear strength tests and micro-hardness measurements were done to evaluate the mechanical properties of different samples. For the shear strength measurement, STM-50 Universal Testing Machine (Santam Co.) was used. Shear strength test was performed under compression by the speed of 0.5mm/min after placement of sample in the shear strength fixture. Microhardness measurement of the joints was carried out with the MHT-10 microhardness device (Anton Paar Company).

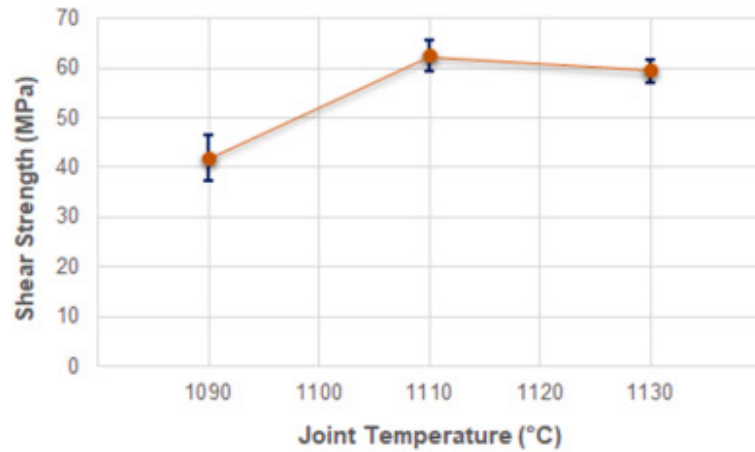
3. Results

Based on the maximum force of the shear test, the shear strength of the samples was calculated at different temperatures. [Table 2](#) displays the bonding temperature, bonding time, measured force values, and calculated shear strength for each joint. [Fig. 2](#) demonstrates the comparison graph for the changes in the shear strength of the bonds over the bonding temperature for the B₄C/BNi-2/B₄C system. [Fig. 2](#) indicates that by increasing the bonding temperature from 1090°C to 1110°C, the shear strength of the bond rises significantly. However, by increasing the bonding temperature from 1110°C to 1130°C, a decrease in the shear strength of the bond is observed. On the other hand, the reason that the data related to the 1150°C sample (S4) is not available in [Table 2](#) and [Fig. 2](#) is that at this temperature, the molten BNi-2 interlayer flowed between the B₄C parts and effused on the brazing fixture, causing the sliding of B₄C pieces on each other (the alignment of B₄C parts on top of each other was disarranged). Therefore no proper bond was achieved at this temperature. [Fig. 3](#) shows the failed samples at 1150°C and the outflow of the molten interlayer on the brazing fixture in two different views.

Table 2. Bonding temperature, bonding time, measured force values, and calculated shear strength values for each brazed specimen with BNi-2 interlayer.

Specimen Code	Bonding Temperature (°C)	Bonding Time (Hour)	Shear Force (kN)	Shear Strength (MPa)
S1	1090	1/2	4.2±0.5	42±5

Specimen Code	Bonding Temperature (°C)	Bonding Time (Hour)	Shear Force (kN)	Shear Strength (MPa)
S2	1110	1/2	6.2±0.3	62±3
S3	1130	1/2	5.9±0.2	59±2
S4	1150	1/2	–	–



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Fig. 2. Comparative graph for the changes in shear strength of the bond over the bonding temperature for the B₄C/BNi-2/B₄C system.



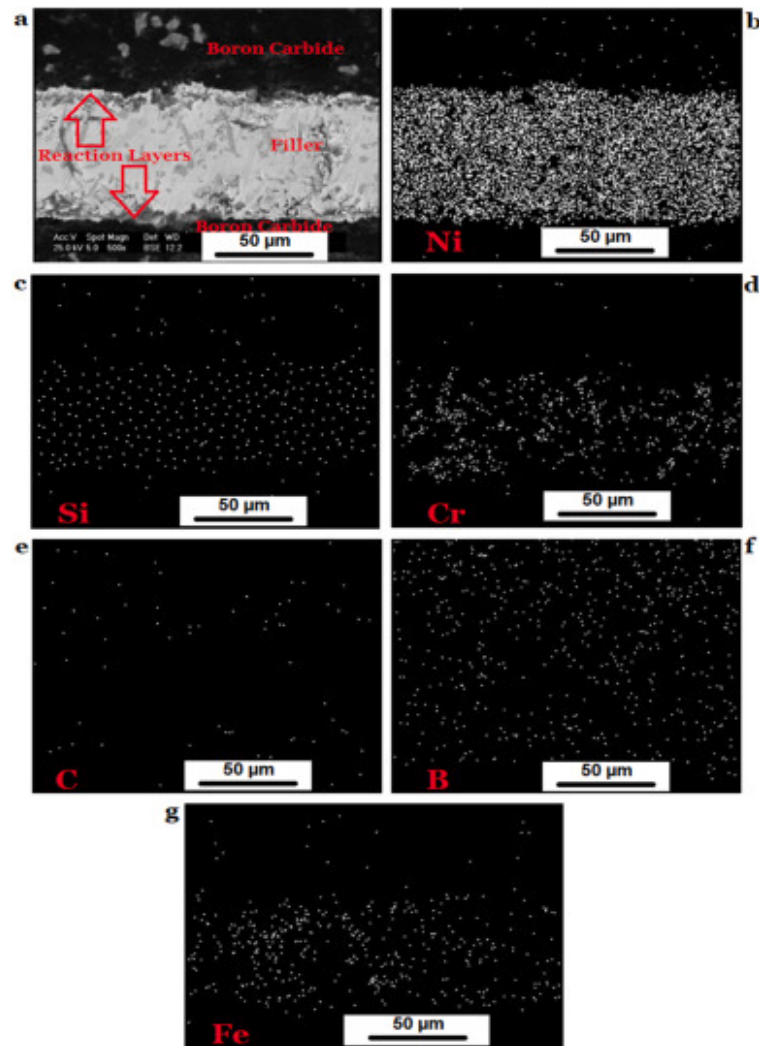
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Fig. 3. Failed bonds at 1150°C and effusion of molten interlayer on the brazing fixture in two different views.

Fig. 4, Fig. 5 display WDS map analyses at the cross sections of S1 and S2 samples (samples with the lowest and highest shear strength values), respectively. Investigation of the SE micrographs related to the cross sections of the two samples shows that the width of the formed reaction layers in the S2 sample has expanded compared to the S1 sample (due to the same magnification of the images). It also seems that, about the S1 sample, the continuity of the reaction layer in the interface is not observed. In other words, at some points in the boundary between the B₄C layer and the interlayer, no specific reaction layer has been formed, and it seems that the reaction layer has been creating in these regions, but its reactions have not fully happened. Another critical point is forming one reaction layer on each side of the bond in both samples. In both samples, the interlayer has been divided into two dark and bright phases (due to the reactions during the brazing process), and it seems that the dark phase is distributed within the bright phase. Concerning each reaction layer, it appears that they are also a combination of bright and dark phases. Regarding the WDS map analyses presented for the main elements of the BNi-2 interlayer (Nickel, Chromium, Silicon, and Iron elements), the diffusion of these elements from the middle layer (interlayer) to the sides of the bond (B₄C layers) is observed in both samples. Comparing the range of diffusion of these elements in the S1 and S2 samples indicates that in the S1 sample

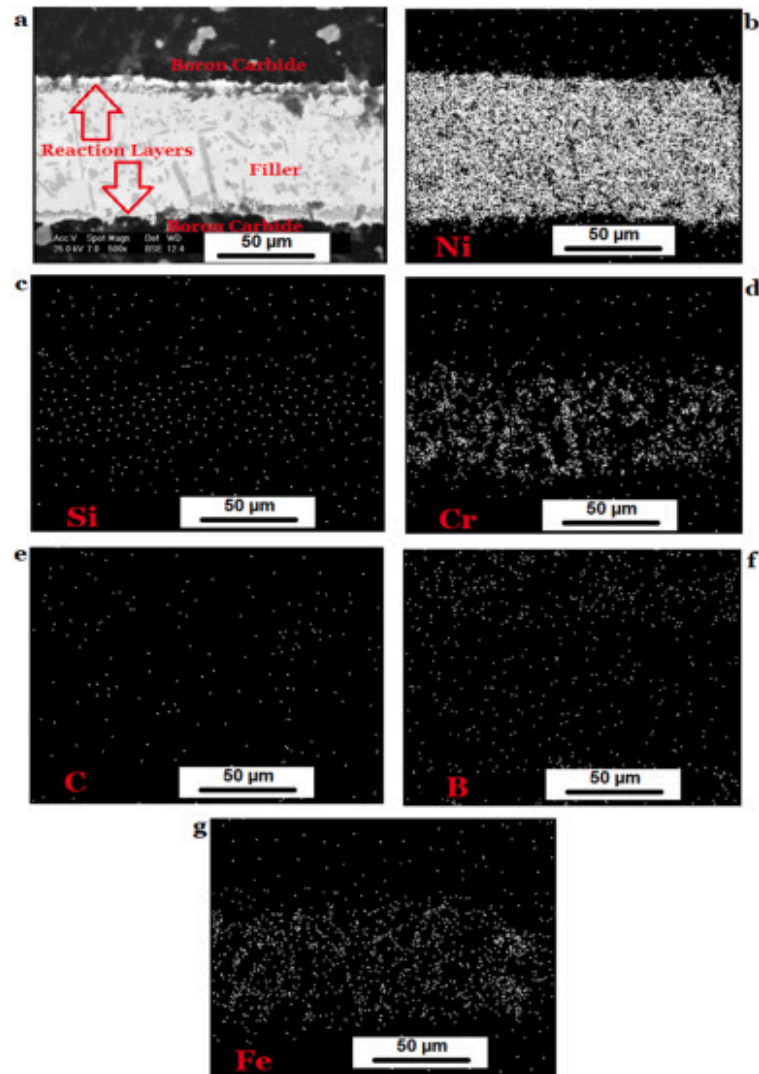
(sample with a lower bonding temperature), the range of diffusion of these elements is chiefly restricted to the reaction layers on the sides of the bond (areas with different colors than the base and interlayer). Small amounts of these elements enter the B₄C ceramic layers on the sides of the bond. However, it looks that the Si element, due to its smaller atomic radius and higher reactivity than other elements (Ni, Cr, and Fe), had more diffusion toward the base ceramic on the sides of the S1 sample. About the diffusion pattern of these elements in the S2 sample (sample with a higher bonding temperature), the increase in the amount and range of diffusion by increasing the bonding temperature is quite apparent. The WDS map analysis related to the C element in the S1 sample shows the limited diffusion of this element into the middle layer (interlayer), and its diffusion is mainly restricted to the reaction layer. The lower bonding temperature associated with this sample seems not to be sufficient to create the needed conditions for broader diffusion of this element. On the other hand, in the analysis related to the B element in the S1 sample, the presence of this element in the middle layer (interlayer) is much more than that of the C element. The WDS map analysis of carbon and boron elements for the S2 sample indicates that by increasing the bonding temperature, the diffusion of boron and carbon elements from the base ceramic layers to the center of the bond (interlayer) has increased.



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Fig. 4. a) The SE micrograph of the interface cross section of S1, b) nickel, c) chromium, d) silicon, e) carbon, f) boron, and g) iron elemental map analyses. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

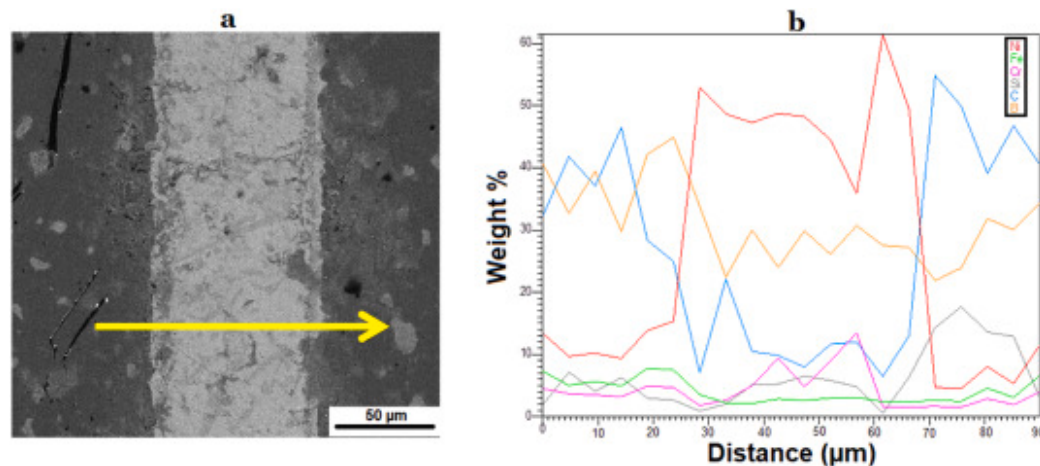


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Fig. 5. a) The SE micrograph of the interface cross section of S2, b) nickel, c) chromium, d) silicon, e) carbon, f) boron, and g) iron elemental map analyses. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

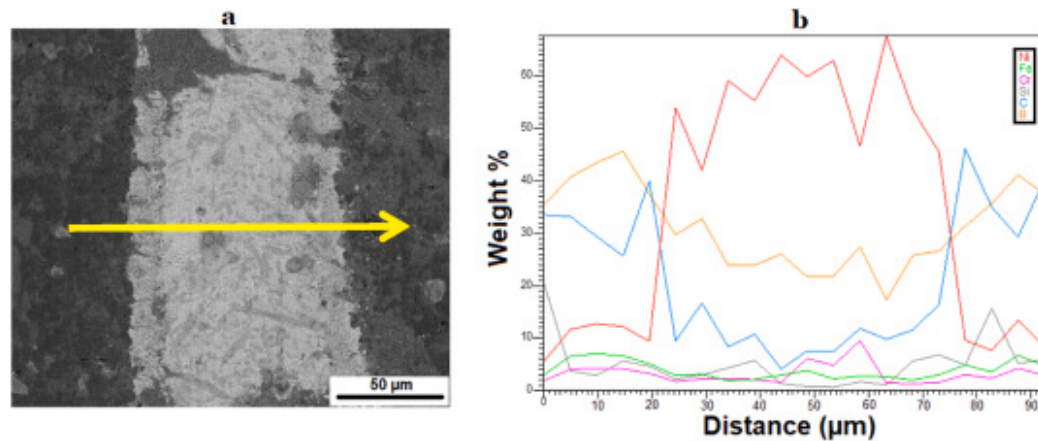
Fig. 6, Fig. 7 illustrate the results of line elemental analysis at the cross-sections of S2 and S3 samples, respectively. These results align well with the findings from WDS map analyses conducted on the samples. An essential observation is the confirmation of the relative diffusion of B and C elements from the bond's sides toward the center, supported by line elemental analysis. Furthermore, the line elemental analysis confirms higher concentrations of B compared to C in the middle parts of the bond interface. SE micrographs of the cross-sections reveal that the interlayer area is predominantly divided into two phases: a bright phase and a dark phase. Considering that Ni and Cr have the highest wt.% shares in the BNi-2 interlayer's chemical composition, the reverse pattern of troughs and peaks of these elements in the interlayer region of different samples indicates that the peaks of each element in the interlayer area are associated with the rich phase of the same element. The harmonic pattern observed in the troughs and peaks of Cr and Fe elements in the reaction regions as well as the diffused sections in the B₄C ceramic suggests a similar diffusional behavior for these two elements. Additionally, the reverse peaks and troughs of Cr and Si elements in the interlayer area confirm the solubility of silicon in the chromium-rich phase. As the Cr content increases, there is an observable rise in the reactivity of Si in the molten alloy, as indicated by the harmonious pattern.



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Fig. 6. a) The SE micrograph related to the cross section of the interface of the S2 sample, b) Line elemental analysis at the cross section of S2 sample.



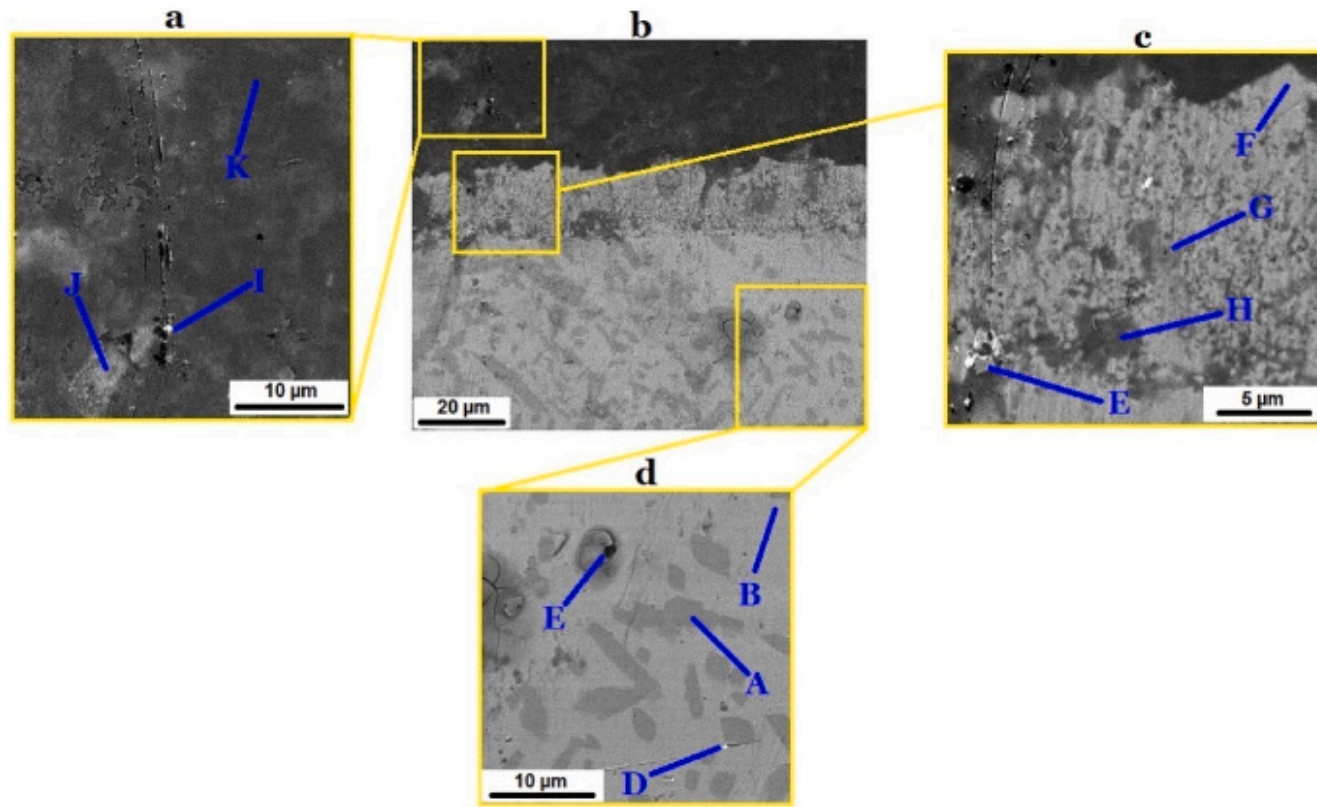
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Fig. 7. a) The SE micrograph related to the cross section of the interface of the S3 sample, b) Line elemental analysis at the cross section of S3 sample.

SE micrographs depicting different regions of the interface of the bonds for S2 and S3 samples have been presented in Fig. 8, Fig. 9, respectively. The elemental analysis results for specified points on these micrographs have been detailed in Table 3. In the interlayer area of both samples, the elemental analyses and examination of formed phases indicate the presence of two main phases. One phase is rich in Ni and nearly devoid of Cr (bright phase - points B and M), while the other phase contains significant amounts of Cr (dark phase - points A and L). These results align with the findings from line and map elemental analyses. Any changes in the color of the formed phases compared to the dominant phases in the interlayer area are attributed to the diffusion of B and C elements into these regions. This matter is evident in the analyses for both samples at points C, D, and N. Also, about the analyses of points C and D in the S2 sample, despite the apparent color difference between these two points (one is quite bright and the other is quite dark), the obtained analyses are similar with little differences, and probably this color change is due to the formation of different phases in these points. Silicon appears to be primarily present in the nickel-rich phase suggesting the probable formation of various nickel silicide compounds in the bright phases of the interlayer area (points B and M). However, in the case of the diffusion of C and B elements into these regions, other potential compounds, such as nickel boride, silicon boride, and silicon carbide, may also form (points C, D, and N). In Cr-rich phases, partial amounts of Si are observed (points A, C, D, H, L, P, and Q), indicating that the availability of chromium increases silicon reactivity in the molten alloy. Due to

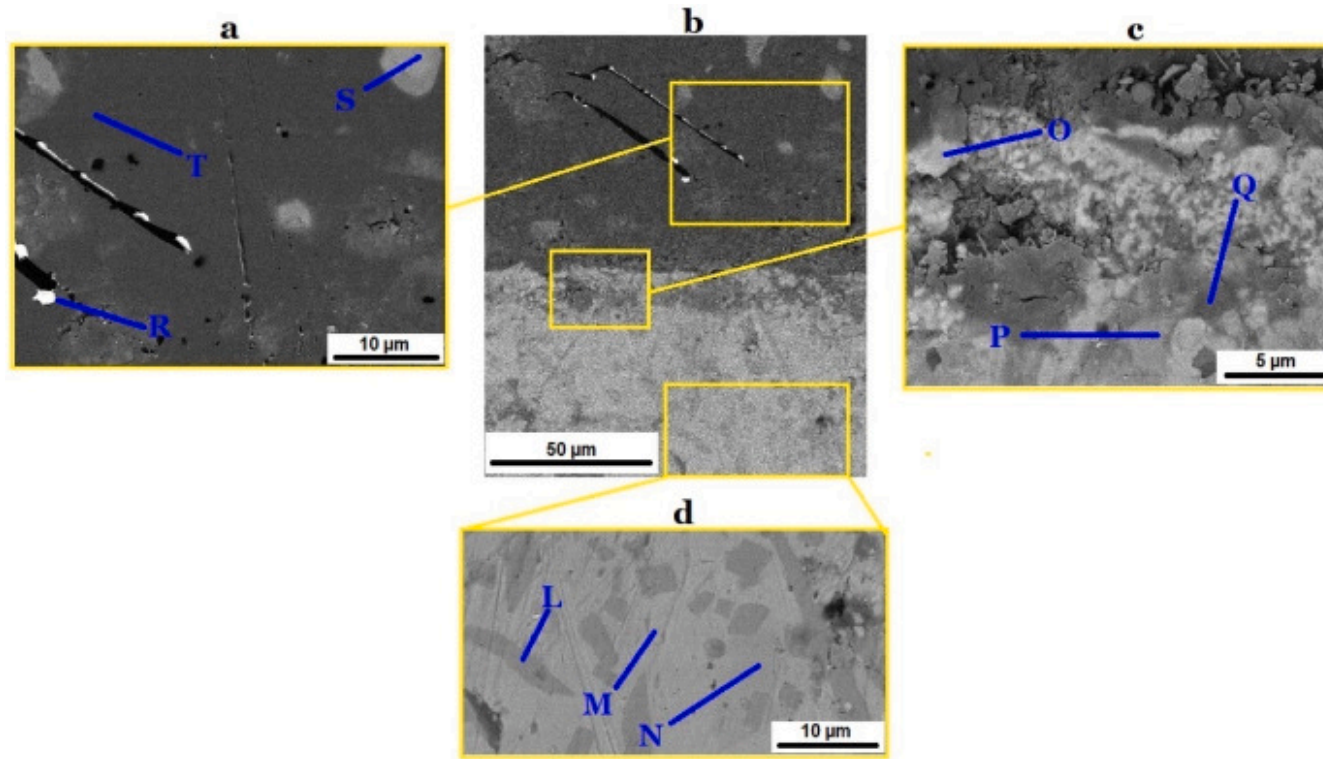
the relatively similar diffusional behavior of iron and chromium, noticeable amounts of iron are present in the chromium-rich phase (points A, C, D, H, L, P, and Q). If B and C elements diffuse into these regions, potential compounds including Fe and Cr borides as well as their carbides are formed (points C, D, and N). Analyzing point E which is observed as a bright spot in the reaction area of the S2 sample reveals some amount of Al and Ti. The presence of these elements is probably due to the presence of sintering-aid compounds in the fabrication process of the B₄C substrate. The diffusion of available elements in B₄C sintering-aid compounds in some areas of the bond interface has caused the formation of phases with relatively distinct colors. Regarding point F which is identified as a bright phase in the reaction region of the S2 sample, the elemental analysis is similar to that of the bright phase in the interlayer area. It is likely that this phase mainly consists of nickel silicide compounds. The analysis of point G which is a relatively dark spot in the reaction region of the S2 sample indicates similarity to the dark phase observed in the interlayer area. Analyzing point H suggests that the slight discoloration compared to point G is mainly due to the diffusion of boron and carbon elements into this region. However, small amounts of Al and Ti may also contribute to the observed color difference. About the analyzed points in the B₄C substrate area, the color change in points I and J compared to the primary color of B₄C is related to the availability of some elements in the interlayer as well as Al and Ti. The elemental analysis of point K indicates the chemical composition of B₄C with minor amounts of diffused elements from the interlayer and small amounts of Al and Ti. Concerning the reaction layer of S3 (points O, P, and Q), the elemental analyses are similar to the corresponding points in the reaction layer of S2 (points F, G, and H). Analyzed points in the ceramic substrate area of S3 (points R, S, and T) exhibit elemental compositions akin to the corresponding points in the ceramic layer of S2 (points I, J, and K). However, differences in the weight percentages of analyzed elements in corresponding points can be attributed to applying various temperatures for brazing, influencing the diffusion pattern of elements.



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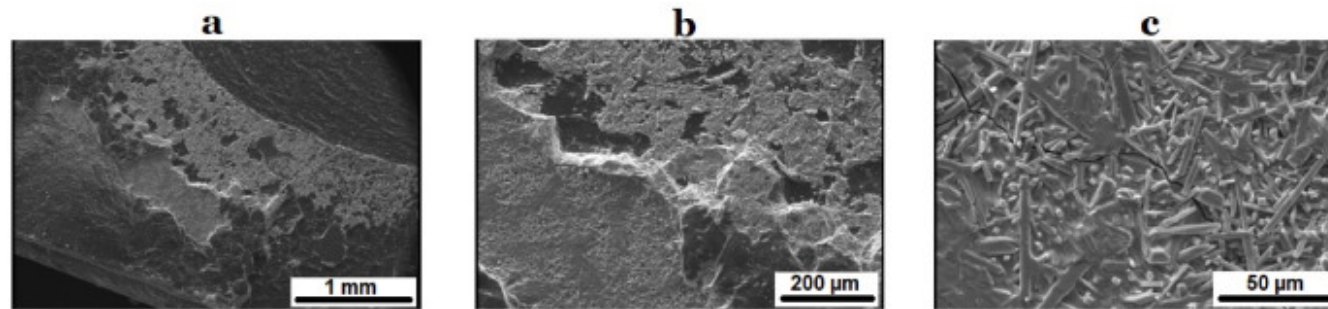
Fig. 8. SE micrographs related to S2 sample: a) B₄C base ceramic, b) cross-section of the interface, c) reaction layer, and d) interlayer.



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Fig. 9. SE micrographs related to S3 sample: a) B₄C base ceramic, b) cross-section of the interface, c) reaction layer, and d) interlayer.



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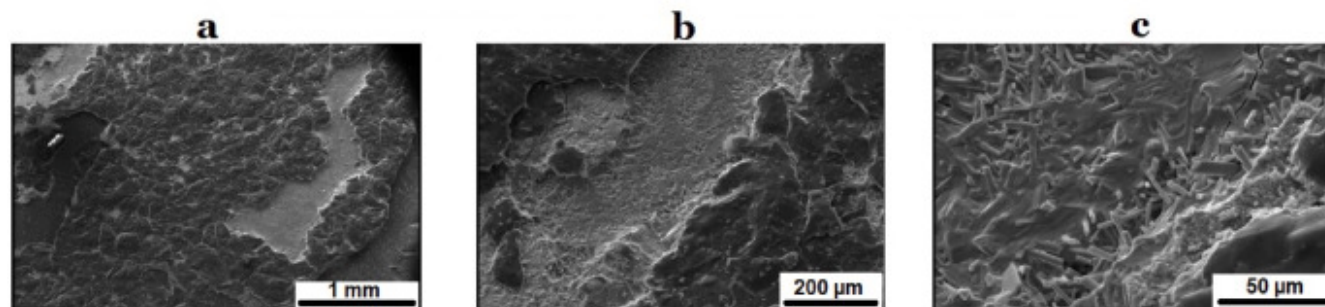
Fig. 10. Fracture surface of S2 sample in three different magnifications.

Table 3. Results of elemental analysis for the specified points on [Fig. 9](#), [Fig. 10](#) (wt. %).

	Ni	Cr	Si	Fe	B	C	Ti	Al
A	42.7	44.8	0.6	11.9	–	–	–	–
B	84.2	4.1	10.8	0.9	–	–	–	–
C	41.1	31.7	0.4	8.5	10.2	8.1	–	–
D	42.3	29.8	0.3	7.8	11.9	7.9	–	–
E	40.3	1.2	4.3	1.4	14.8	18.1	14.2	5.7
F	76.2	1.4	20.6	1.8	–	–	–	–
G	43.6	44.8	0.4	11.2	–	–	–	–
H	42.1	26.2	1.8	4.7	10.2	12.2	0.2	2.6
I	1.1	0.4	17.7	0.5	51.3	13.9	1.9	13.2
J	1.7	1.1	4.4	1.3	31.2	11.6	46.1	2.6

	Ni	Cr	Si	Fe	B	C	Ti	Al
K	1.2	0.6	0.8	0.8	74.3	20.6	0.4	1.3
L	49.3	41.6	0.8	8.3	–	–	–	–
M	78.9	5.5	14.2	1.4	–	–	–	–
N	58.6	10.1	0.9	13.8	5.3	11.3	–	–
O	76.2	0.5	23.1	0.2	–	–	–	–
P	51.8	22.6	0.8	5.7	12.1	6.2	0.3	0.5
Q	45.9	26.6	1.1	5.1	8.2	12.1	0.5	0.5
R	1.2	1.1	7.1	1.4	39.7	24.9	21.8	2.8
S	2.9	0.6	3.1	0.7	33.6	20.2	31.2	7.7
T	1.8	1.7	1.1	0.9	71.5	21.6	0.6	0.8

[Fig. 10](#), [Fig. 11](#) present Scanning Electron (SE) Micrographs of the fracture surfaces for S2 and S3 samples, respectively. In these images, the dark-colored areas represent sections where the fracture surface has traversed through the ceramic, resulting in the removal of some ceramic parts. The passage of the fracture surface through the ceramic may contribute to the higher bond strength in these areas compared to the strength of the adjacent ceramic. This effect may be associated with the weakening of these sections of the ceramic due to the development of residual stresses caused by the CTE mismatch between the ceramic and the metallic components of the bond.

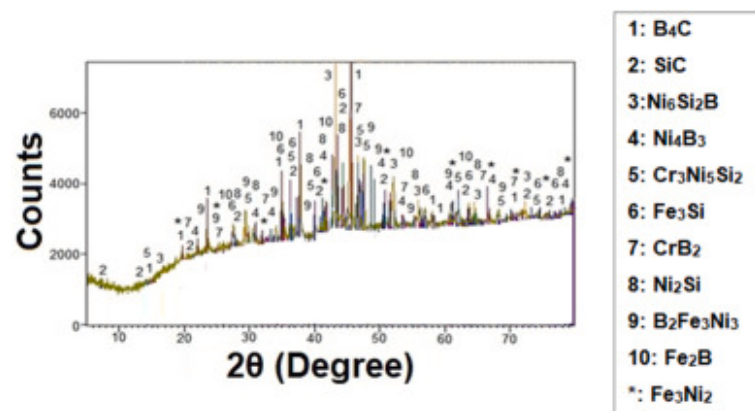


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Fig. 11. Fracture surface of S3 sample in three different magnifications.

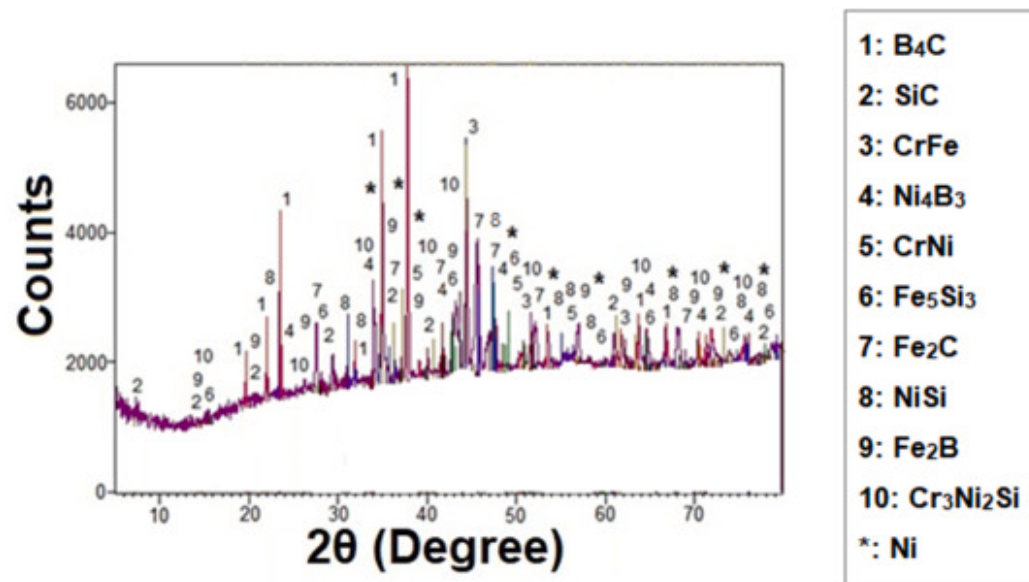
Fig. 12, Fig. 13 display the XRD spectra for the fracture surfaces of S2 and S3 samples, respectively. The compounds and phases identified based on the XRD analyses are in good agreement with the elemental analyses of the system. Upon comparing the identified phases and compounds at the fracture surfaces of the S2 and S3 samples, it is clear that the sufficiently high process temperature causes the decomposition of the base ceramic and leads to a variety of interfacial reactions between the main elements of the molten interlayer and C and B elements. Additionally, various reactions between the elements of the molten interlayer are evident.



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Fig. 12. XRD analysis at the fracture surface of the S2 sample.

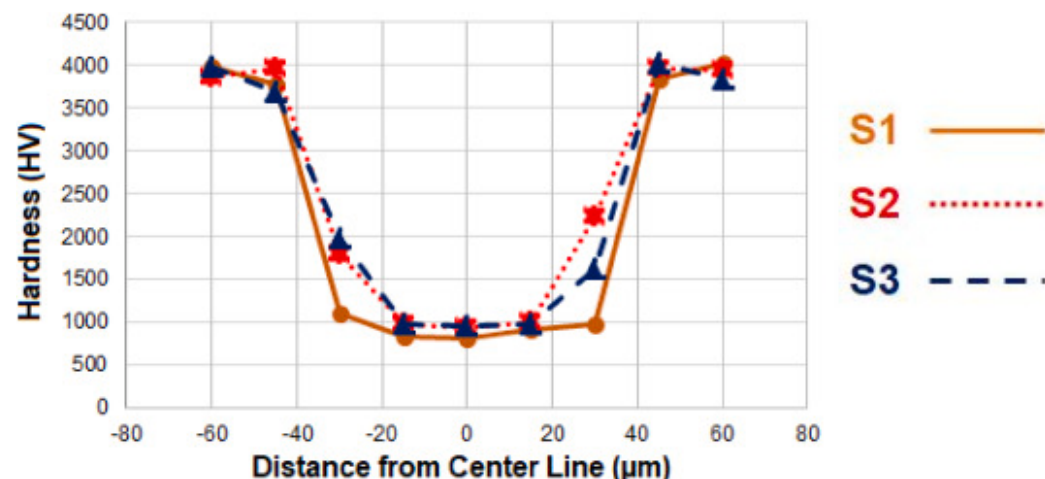


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Fig. 13. XRD analysis at the fracture surface of the S3 sample.

Fig. 14 presents a comparison graph of the bond hardness for S1, S2, and S3 samples. The results indicate that as one moves from the bond center to the sides of the bond in all three samples, the hardness gradually increases. Additionally, the hardness values measured at the three middle points of the S1 sample are lower than those of the S2 and S3 samples. As one moves away from the middle part of the bond and enters the range of the reaction layer, there is an increase in hardness in all three samples. This increase is more pronounced in the S2 and S3 samples compared to the S1 sample. The variation in the measured hardness values in different samples, especially in the reaction regions, makes it challenging to attribute these values to a specific phase. Notably, all three samples exhibit a significant increase in measured hardness values upon entering the B₄C ceramic layer.



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Fig. 14. Comparison diagram of bond hardness for S1, S2, and S3 samples.

4. Discussion

The results in Table 2 and Fig. 2, Fig. 3 suggest that an increase in the bonding temperature generally enhances the fluidity of the molten interlayer, facilitating its easier diffusion into the porosities of the ceramic member. This elevated bonding temperature also promotes increased interactions between ceramics and the molten interlayer, creating favorable conditions for the diffusion of disposed elements [11,12]. Therefore, it can be inferred that the heightened fluidity of the molten interlayer and enhanced interactions in the ceramic-metal system due to the increased bond temperature are the primary reasons for achieving bonds with higher shear strength. The higher shear strength of the S2 sample compared to the S1 sample can be attributed to these factors. Another crucial consideration in explaining the effect of the bonding temperature parameter on the shear strength of the sample is that an excessive increase in bond temperature may lead to excessive fluidity of molten filler alloy and increased destructive interactions in the ceramic-metal system, resulting in a loss of shear strength in the bond [12]. The observed decrease in shear strength of the S3 sample compared to the S2 sample could be mainly attributed to an increase in destructive interactions within the ceramic-metal system. The failure to achieve the desired bonding at 1150°C (S4 sample) may be because at this temperature the bonding temperature approaches the upper limit of the recommended temperature range for brazing of

BNi-2 interlayer, as per the AWS standard (the upper limit is 1177°C) [20]. The excessive fluidity of the molten alloy at this temperature may lead to the sliding of bonding components on each other and the outflow of the molten alloy onto the brazing fixture.

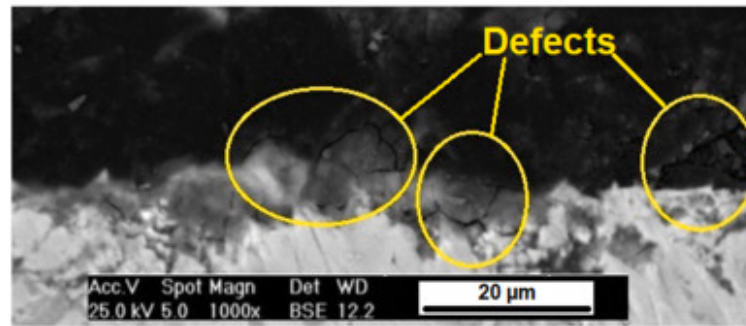
Indeed, in numerous brazing processes, minimizing the temperature is often desirable for several reasons. These include conserving heat energy, reducing the thermal impact of the process, mitigating reactions between the interlayer and the base material, and extending the life of equipment and tools [22]. Achieving a balance between lowering process temperatures and maintaining the quality of the bond is crucial for optimizing efficiency and effectiveness in brazing applications. This delicate balance ensures that energy is used efficiently, equipment longevity is preserved, and the brazed joints meet the required quality standards.

The mechanical properties and strength of brazing bonds are significantly influenced by the structure of the interface and the residual stresses developed due to the CTE mismatch between bond components. CTE mismatch between the bonded materials results in thermal residual stresses during the cooling process, which can substantially affect bond strength [19]. As explained earlier, an increase in bonding temperature can exacerbate the impact of CTE mismatch on bond strength. The decline in strength observed in the S3 sample compared to the S2 sample (1130°C) might be attributed to the higher bonding temperature, leading to increased thermal residual stresses. The presence of ceramic and metal components in the bond introduces complexities regarding CTE mismatch. However, the relatively high nickel content in the BNi-2 interlayer provides sufficient ductility, allowing it to deform plastically under the induced stresses. This plastic deformation helps to alleviate the impact of destructive stresses to some extent, contributing to the overall proper strength of the bonds. The appropriate ductility of the BNi-2 alloy enables it to maintain these properties even at relatively higher temperatures, presenting a notable advantage compared to other interlayers.

The study demonstrates promising results, with the highest shear strength of 62MPa for the brazed sample using the BNi-2 interlayer at 1110°C. This shear strength surpasses reported values for other filler alloys like Ticusil, Cu-4.5at.%Ti, and several instances of Cusil-ABA. The BNi-2 alloy's notable corrosion resistance and high-temperature strength contribute to its prioritized use in numerous applications, even when its obtained shear strength is relatively lower compared to other fillers such as Cu-9 at%B and certain reported cases for Cusil-ABA [11,12,19].

The observations from Fig. 4, Fig. 5 suggest several factors influencing the width and continuity of the reaction layer in different samples. The increase in the width of the reaction layer in S2 compared to S1 can be attributed to the higher bonding

temperature, resulting in increased fluidity of the molten alloy and enhanced system interactions. This matter facilitates the diffusion of elements, leading to a broader reaction layer. Conversely, the discontinuity of the reaction layer in the interface of S1 may be due to inadequate bonding temperature, preventing complete interfacial reactions across the entire bond area. The higher amounts of B elements compared to C elements in the reaction layer can be influenced by the lighter weight of B, leading to greater diffusion. Additionally, the presence of B in the chemical composition of the BNi-2 interlayer contributes to the abundance of B in the reaction layer. The high proportion of Ni in the BNi-2 alloy, unable to form carbide compounds [[23], [24], [25]], may lead to boron supply from the decomposition of B₄C in the absence of boron in the interlayer. Therefore, there will be higher tendency for the reaction with B₄C and its disintegration, which consequently leads to the broader diffusion of boron and carbon elements. Nevertheless, in this study, the availability of boron element in the primary composition of BNi-2 alloy prevents the further diffusion of the boron element from the substrate side to the center of the bond. The observed diffusion patterns of elements and shear strength results indicate that the lower strength of the S1 sample can be primarily attributed to insufficient bonding temperature for the completion of interfacial reactions throughout the bonding area, resulting in inadequate bonding at the entire bond area. In the S1 sample, relative diffusion of elements, local widening of the bonding area (leading to a heterogeneous bond interface), and changes in the chemical composition of the interface contribute to irregular shrinkage during the cooling process, leading to defects in the bond interface. On the other hand, for the S3 sample, the relatively high temperature appears to have a more pronounced impact on the CTE mismatch between the ceramic component and the metallic interlayer. This heightened CTE mismatch results in more destructive residual stresses in the bond interface, leading to defects and a loss of bond strength. Additionally, the higher temperature in the S3 sample may cause an overreaction of the interlayer elements with the ceramic due to increased fluidity of the molten interlayer. This overreaction leads to more expansive brittle areas, changes in the chemical composition of the bond interface, and an exacerbation of irregular shrinkage during the cooling process, contributing to the formation of defects. The formation of voids, cracks, and defects in the interface is associated with irregular shrinkage during cooling or the creation of residual stresses due to CTE mismatch. Fig. 15 displays an SE micrograph of the cross-section of the interface of S3, highlighting defects marked with yellow circles. These defects are likely a result of CTE mismatch, and their formation contributes to the decrease in bond strength.



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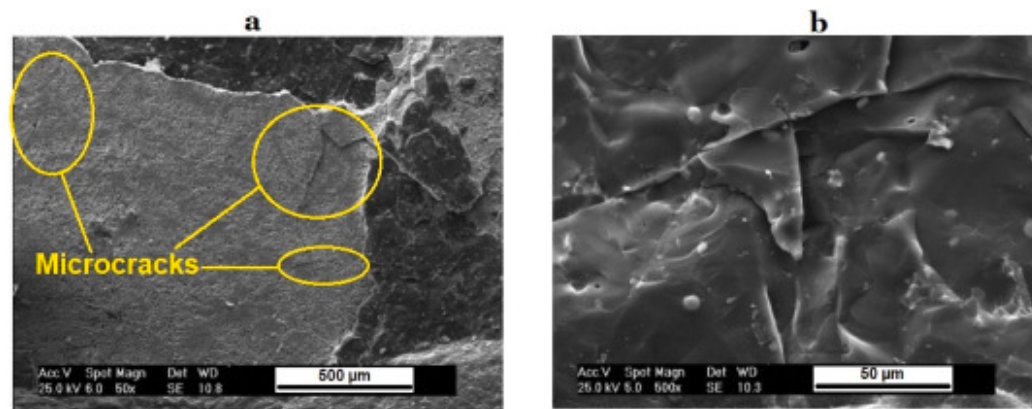
Fig. 15. The SE micrograph of the interface cross-section of the S3 sample and some of the created defects.

The reverse peaks and troughs observed in the line elemental analyses of Si and Cr elements are crucial (Fig. 6, Fig. 7), as these elements can form compounds that improve wettability and, consequently, enhance bond strength. Compounds such as SiC (silicon carbide), chromium boride, and others contribute to improved bonding strength. The reverse behavior of Si and Cr elements indicates the advantage of their simultaneous presence in achieving higher bond strength. Furthermore, the occurrence of asymmetric diffusion of various elements from the center of the bond to the sides of the bond is noteworthy. Asymmetric diffusion can lead to the formation of a heterogeneous interface and, consequently, reduce the strength of the bond.

The observations regarding the fracture surfaces of the samples (Fig. 10, Fig. 11) indicate that the ratios of the removed areas from the ceramic at the entire fracture surface for both samples are similar, which might be attributed to the close shear strength values of these two samples (62 MPa for S2 and 59 MPa for S3). In Fig. 10, Fig. 11c, the presence of surface cracks at the fracture surface of the S2 and S3 samples is notable. The formation of these cracks can be linked to the high temperature of these samples, which enhances the negative impact of CTE mismatch between the ceramic and metal components of the bond as well as increases destructive interactions between the bond components.

Fig. 16a shows the formation of surface cracks at the fracture surface of the S1 sample. One of the significant reasons for these cracks in this sample may be associated with the low temperature, preventing the completion of reactions at the interface. This incomplete reaction leads to changes in the chemical composition of the interface, intensifying irregular shrinkage during the cooling process. The CTE mismatch of the bond components can also contribute to the formation of destructive residual stresses,

increasing the likelihood of surface crack formation. Fig. 16b presents the SE micrograph of the fracture of the B₄C part in the S2 sample. This fracture seems to be influenced by intergranular and transgranular mechanisms, representing a combination mode of brittle fracture [[26], [27], [28]].



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Fig. 16. a) Created surface cracks at the fracture surface of the S1 sample, b) The SE micrographs of the fracture in the B₄C part of the S2 sample.

The comparison of the results from XRD analyses of the fracture surfaces of the samples (Fig. 12, Fig. 13) reveals that in the S2 sample, a more diverse array of reactions between the interlayer elements and carbon and boron elements has occurred. This diversity has led to various compounds such as SiC, Ni₄B₃, CrB₂, Fe₂B, Ni₆Si₂B, and B₂Fe₃Ni₃. These reactions may contribute to the higher strength of the S2 bonding. In contrast, the S3 sample exhibits various reactions between the molten interlayer elements, resulting in the formation of intermetallic compounds such as CrFe and CrNi. Additionally, S3 shows the formation of compounds like SiC, Ni₄B₃, Fe₂C, and Fe₂B through reactions between the interlayer elements and carbon and boron elements. However, the greater variety and more substantial peaks of such compounds in S2, along with the relatively high hardness of many formed compounds in the interface of S2, are likely the main reasons for the higher strength of S2 compared to S3. It's noteworthy that the relatively high hardness of many formed compounds in the interface of S2 is a contributing factor to the improved mechanical properties of this bond. Generally, as the temperature increases, conditions become more favorable for enhanced diffusion of the elements, providing the necessary circumstances for reactions between susceptible elements and

forming various compounds and phases. However, in situations where the bonding temperature is already sufficient for element diffusion, further increases might not significantly impact bonding properties. The comparison of the formed compounds in the interface of the S2 and S3 samples along with their respective effects on the final bond properties supports this explanation. An interesting observation is the presence of the iron carbide phase in the S3 sample. The higher temperature of S3 likely created the conditions necessary for the simultaneous formation of iron carbide and iron boride phases.

The observed increase in hardness towards the sides of the bond is likely attributed to the diffusion of elements and the subsequent formation of new phases with higher hardness in these regions. The BNi-2 interlayer contains a variety of elements such as Ni, Cr, Si, and Fe, which can react with boron and carbon elements in different forms, allowing the formation of various phases and compounds within the bond layers. Due to this complexity, it becomes challenging to precisely attribute the hardness of each point to a particular phase. The reported hardness in the middle layer, represented by the three midpoints of each diagram, looks to be mainly associated with nickel silicide phases. These phases dominate the middle region as a bright color phase, with the dark phase dispersed within it. Higher hardness values in the three middle points of the S2 and S3 samples compared to the S1 sample can be linked to the increased temperature in S2 and S3, promoting the diffusion of base elements toward the bond center, leading to the formation of new phases with higher hardness.

The tangible increase in hardness values in the reaction layers of S2 and S3 samples compared to S1 sample may be related to the lower bonding temperature of S1 (1090°C), which might not be sufficient for completing interfacial reactions. Thus, the formation of hard phases has not been entirely achieved. Overall, the measured hardness values in the range of 1000–3000 HV can be related to the formation of different phases, including nickel borides and silicides, chromium borides and carbides, iron borides and carbides, silicon carbide, and compounds such as Ni₆Si₂B, NiC₃B₁₅, and B₂Fe₃Ni₃.

5. Conclusions

The key findings of this study can be summarized as follows:

1. Successful B₄C/B₄C joint fabrication was achieved at 1090, 1110, and 1130°C with a 1/2h brazing time, with failure at 1150°C due to excessive molten interlayer fluidity.
2. The highest shear strength (62MPa) was at 1110°C.

3. Increased bonding temperature improved interlayer fluidity and bonding, while further increase (1130°C) led to decreased strength due to destructive reactions and CTE mismatch.
4. Fracture surface examination at 1110 and 1130°C showed similar shear strength and compound presence, but 1110°C displayed a greater variety and more substantial peaks, likely contributing to its higher strength.
5. The hardness increase in the reaction layer of 1110 and 1130°C samples suggested incomplete reactions at 1090°C due to its lower bonding temperature.

CRedit authorship contribution statement

A. Amirnasiri: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **S.E. Mirsalehi:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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